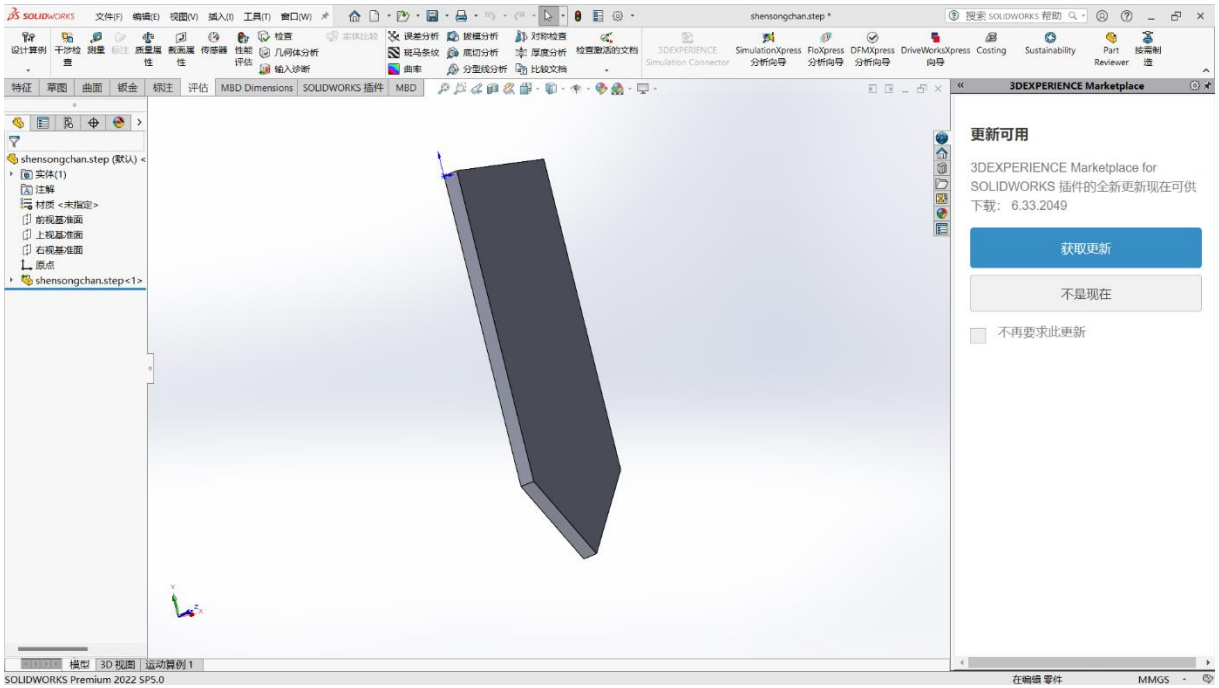
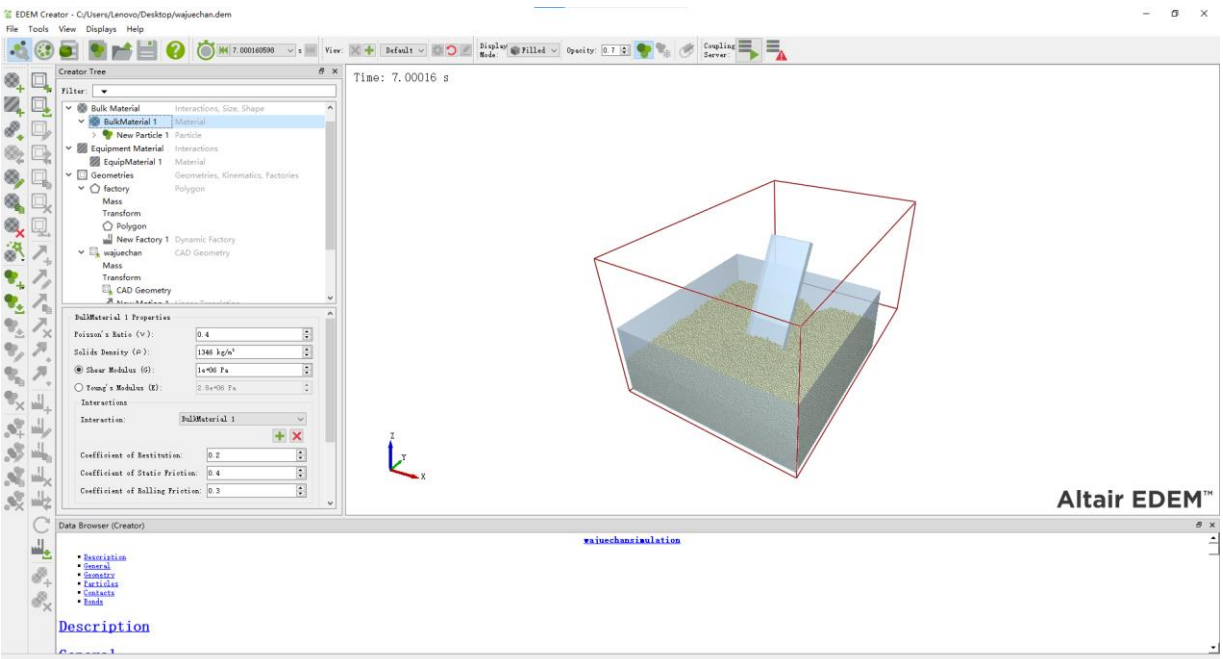
机化 2201　黄钊伟　2022013106

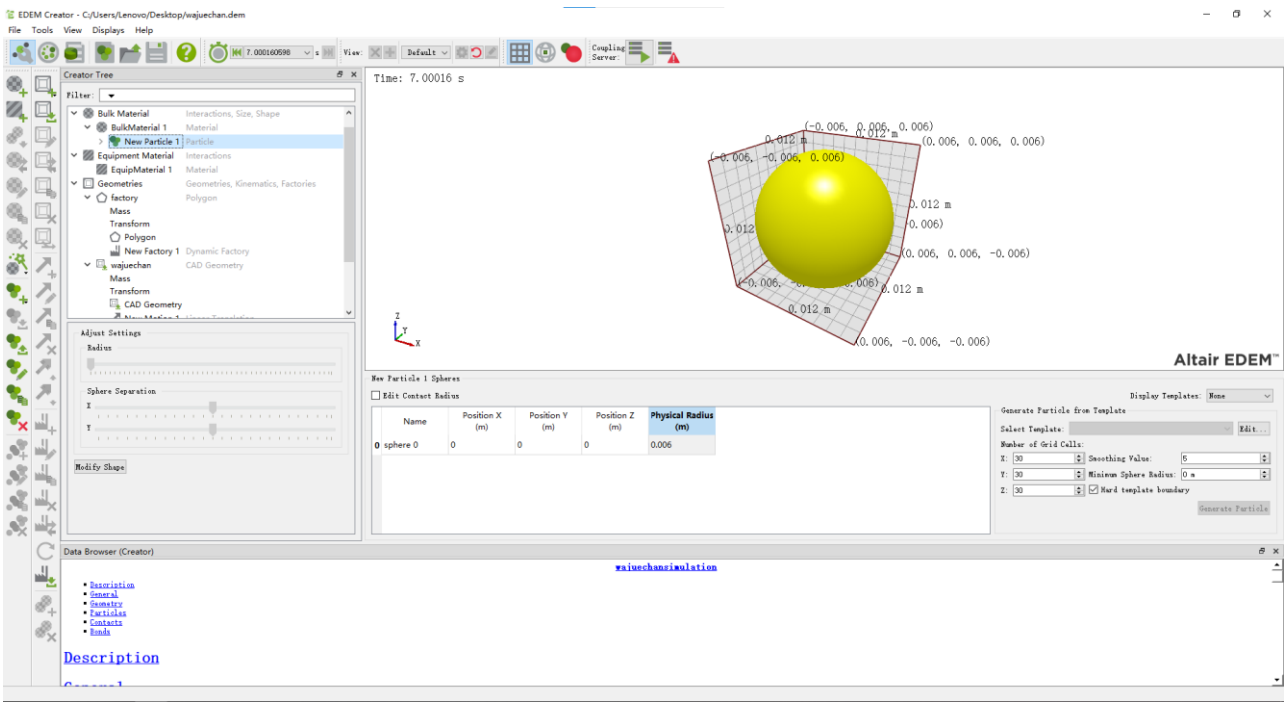
1、挖掘铲三维建模，长度 770mm



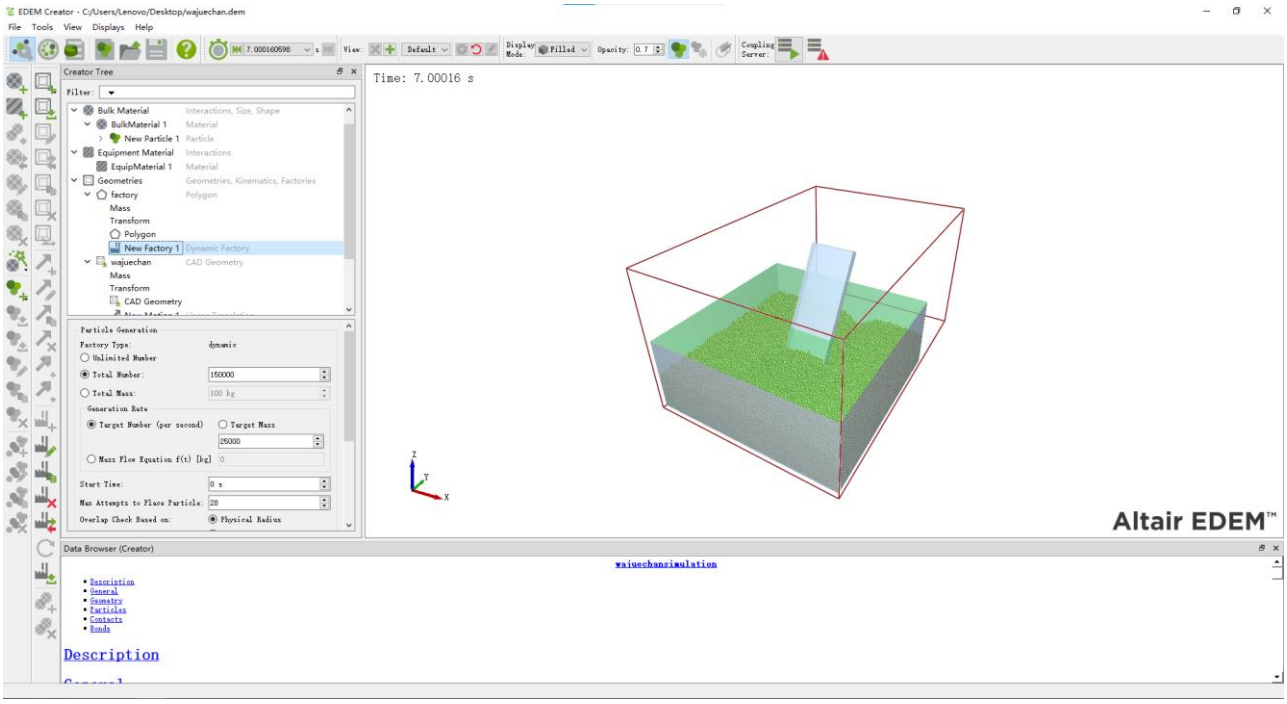
2、颗粒参数设置



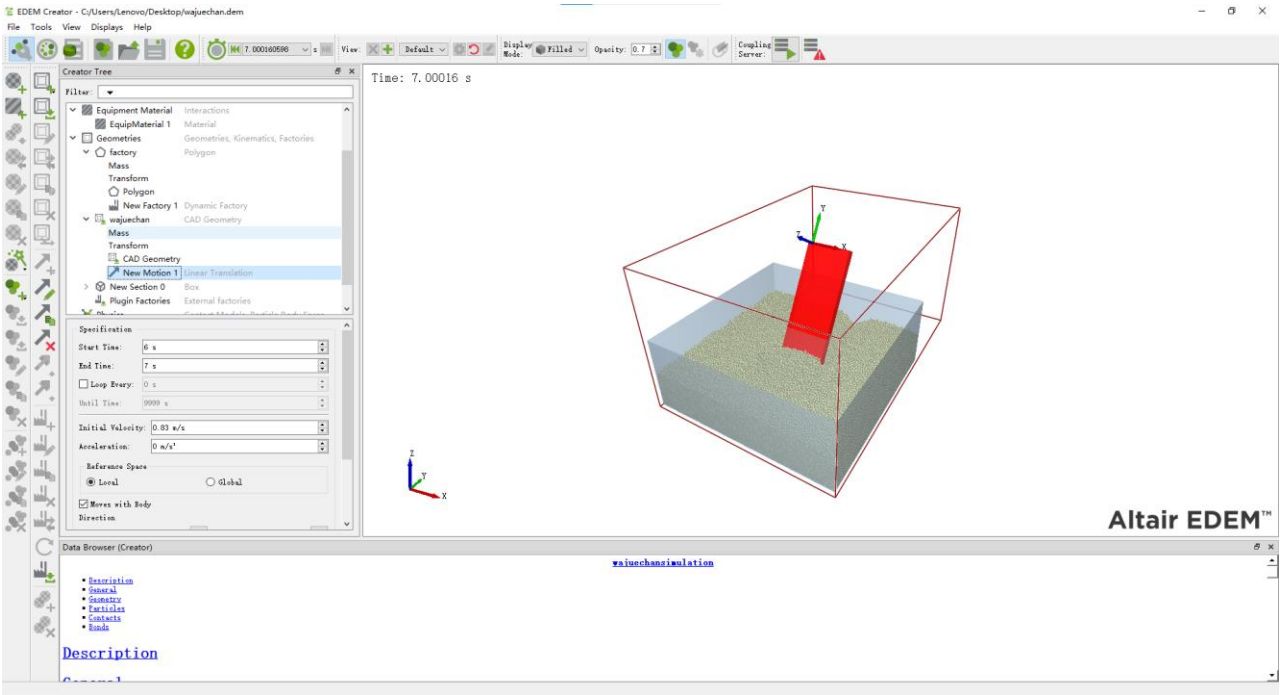
3、颗粒直径为 0.006m



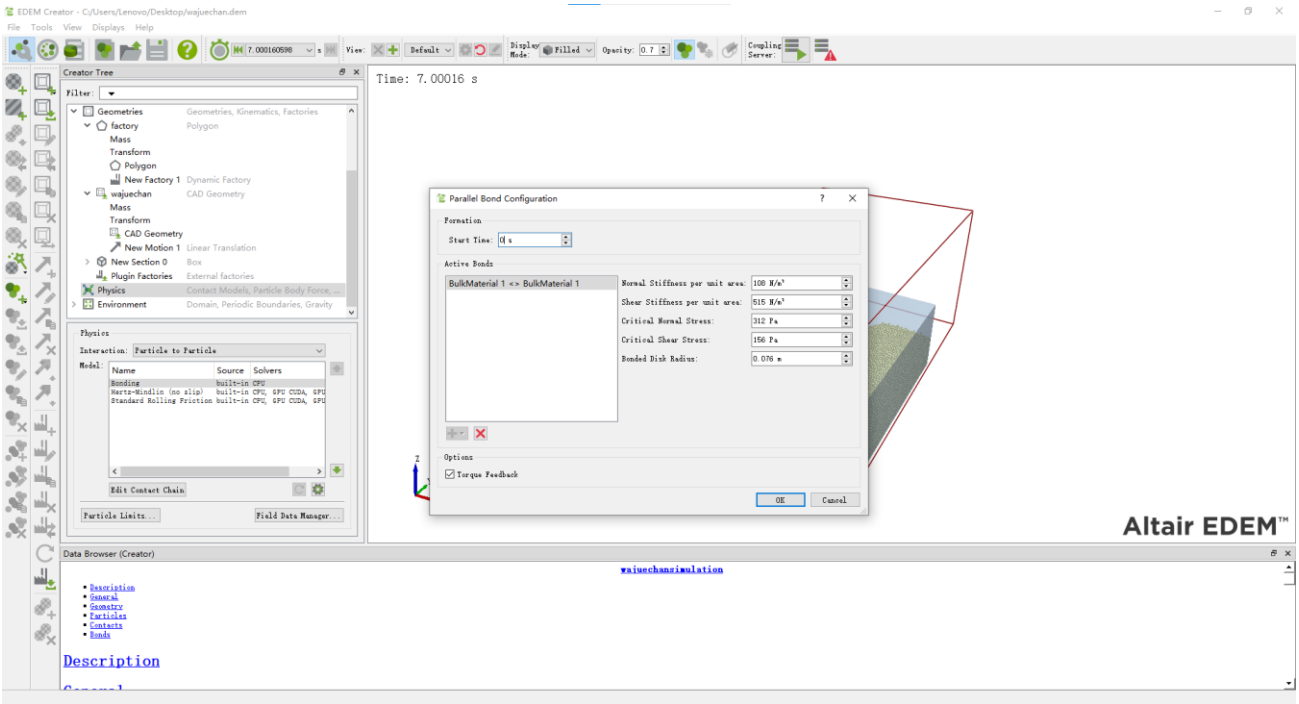
4、颗粒工厂参数设置，总颗粒数为 150000，下落速度 25000 颗/秒



5、挖掘铲速度设置为 0.83m/s，加速度为 0，匀速运动

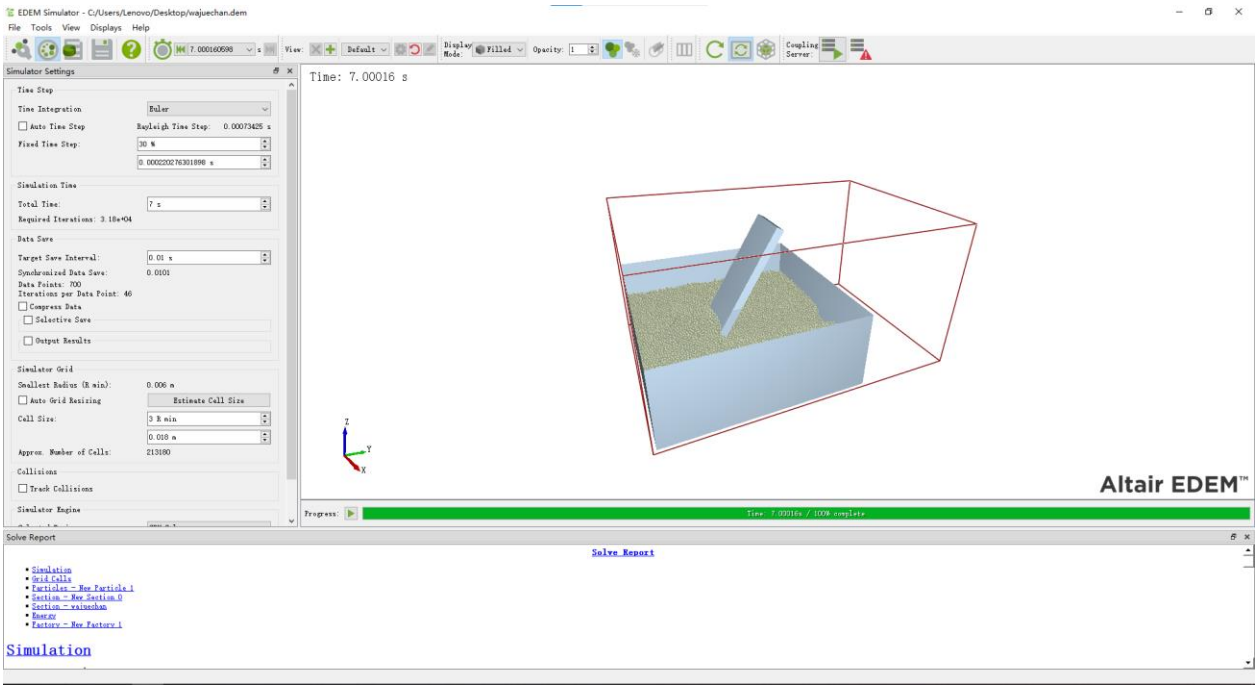


6、接触参数设置

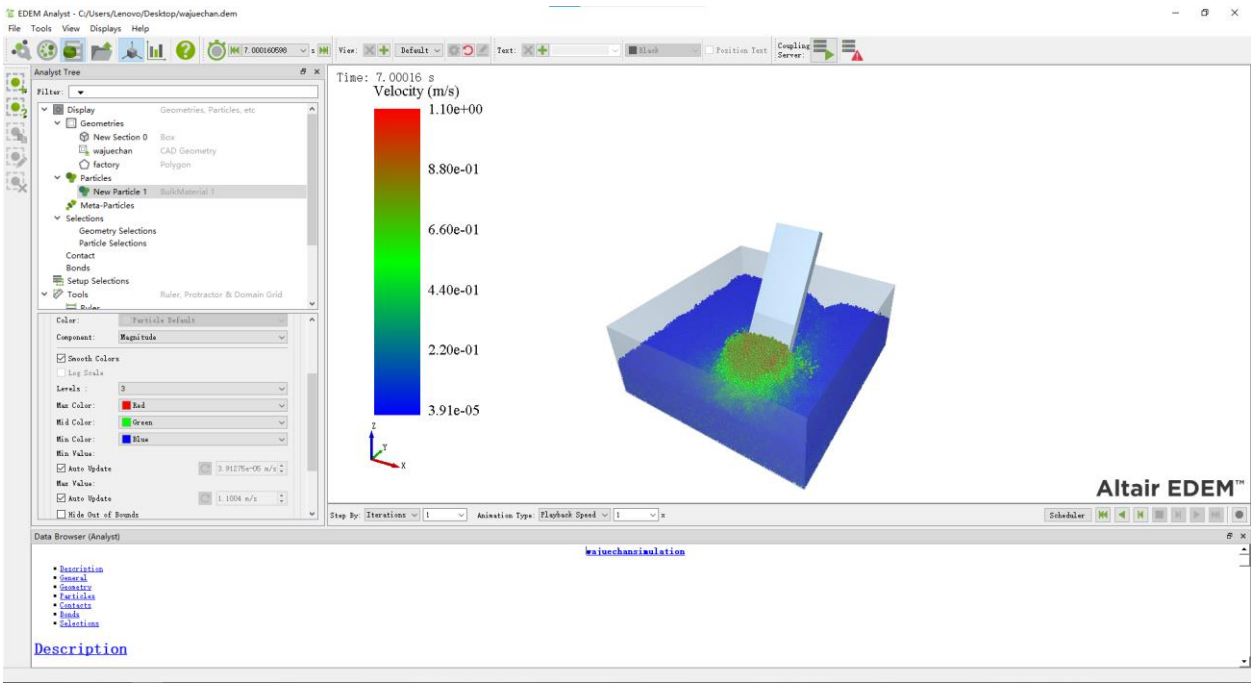


7、时间参数设置 30%，网格参数为 3R，即 0.018m，仿真时间 7s，

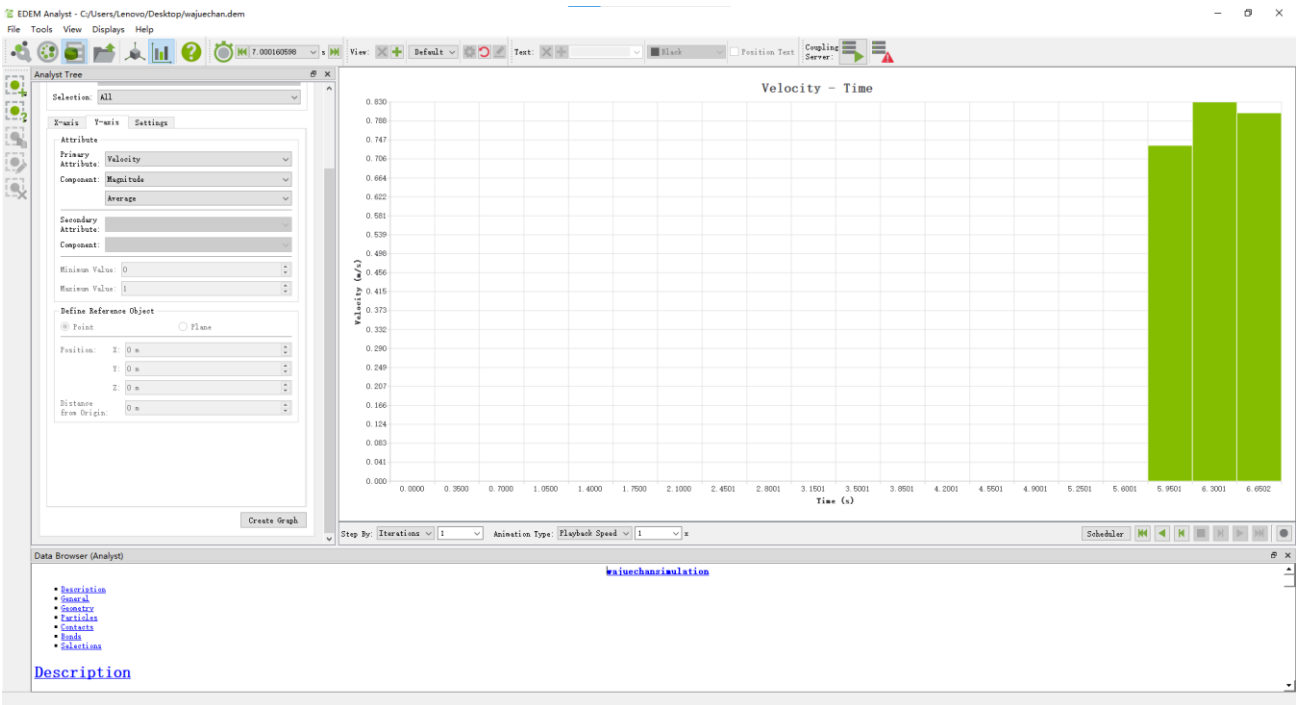
仿真结果如下。



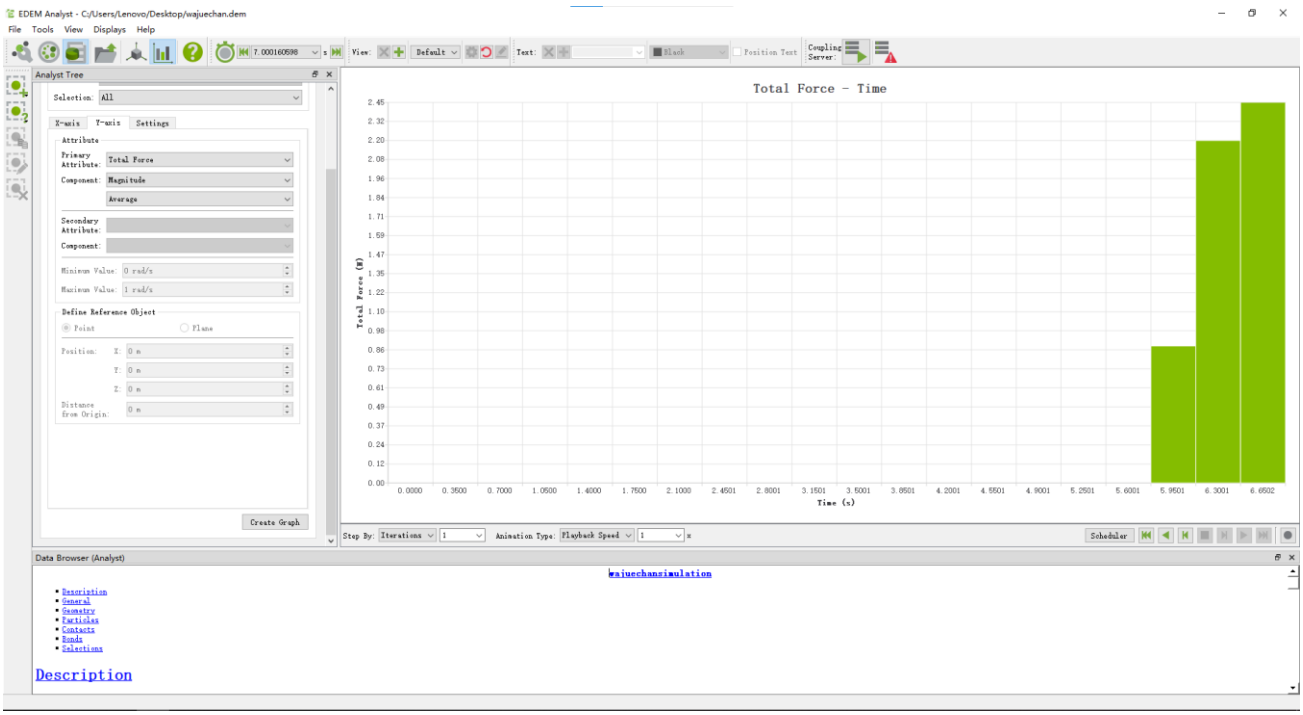
8、土壤颗粒速度云图如图所示，由此可分析土壤扰动情况



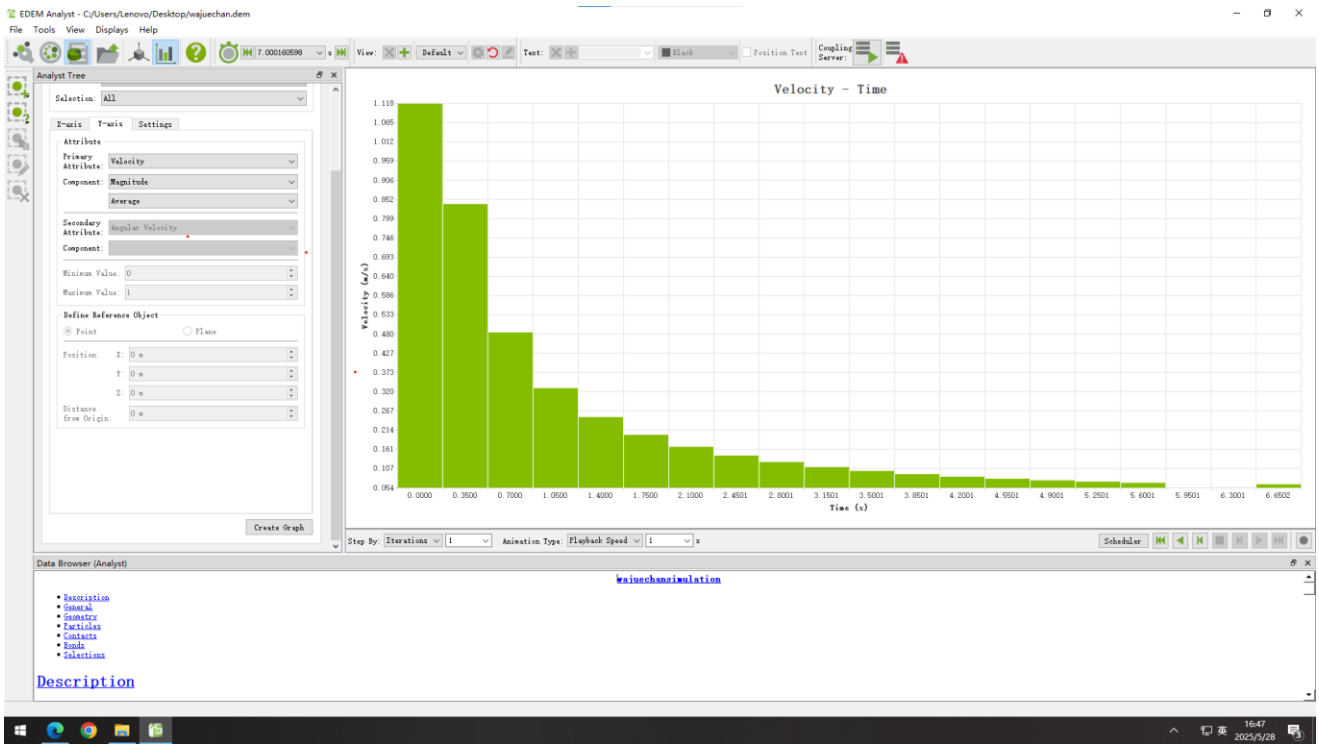
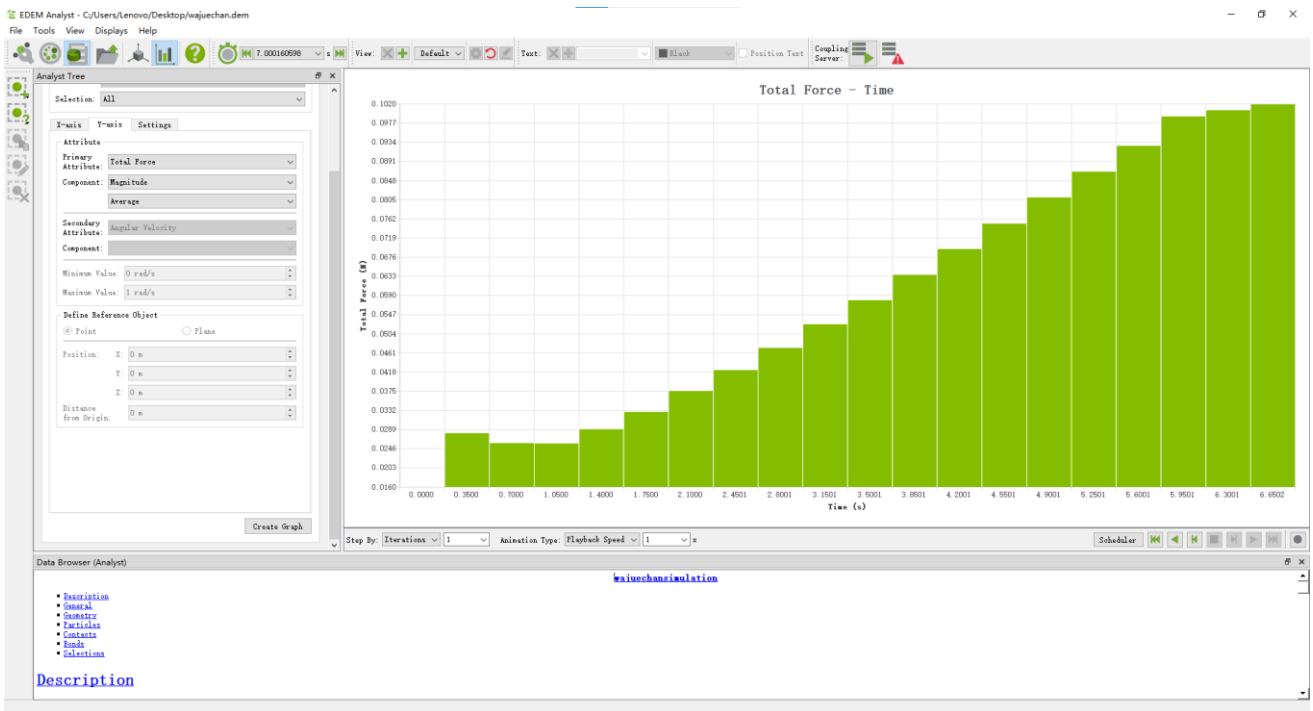
9、挖掘铲运动时间为 6s 到 7s，速度-时间柱状图如图所示



10、挖掘铲运动时间为 6s 到 7s，总压力-时间柱状图如图所示



11、土壤颗粒下落时间为 0s 到 6s，速度-时间柱状图如图所示



12、土壤颗粒下落时间为 0s 到 6s，总压力-时间柱状图如图所示